

Project Report: Unemployment Analysis with Python

Organization: Oasis Infobyte

Author: Bassam Kasar

Tools: Python, Pandas, Matplotlib, Seaborn

1. Executive Summary

This project provides a comprehensive analysis of the unemployment landscape in India, specifically focusing on the period between 2019 and 2020. The primary objective was to quantify and visualize the economic impact of the COVID-19 pandemic on the labor market. By utilizing Python's data science ecosystem, the analysis revealed a dramatic surge in unemployment rates coinciding with the national lockdown, with urban areas experiencing a more significant impact than rural regions.

2. Problem Statement

Unemployment is a critical economic metric, representing the percentage of the labor force that is without work but actively seeking employment. The onset of COVID-19 caused unprecedented disruptions to global economies. This project aims to use data-driven insights to analyze how these disruptions manifested in the Indian context across different states and sectors.

3. Dataset Description

The analysis was performed using the dataset "**Unemployment in India.csv**". The data contains the following key features:

- **Region:** The specific Indian state.
- **Date:** The month and year of the observation.
- **Frequency:** The interval of data collection (Monthly).
- **Estimated Unemployment Rate (%):** The percentage of the labor force unemployed.
- **Estimated Employed:** The count of individuals currently working.

- **Estimated Labour Participation Rate (%):** The percentage of the working-age population that is either working or actively looking for work.
- **Area:** Classification of the region as either **Rural** or **Urban**.

4. Methodology

The project followed a standard Data Science pipeline:

- **Data Preprocessing:**
 - Stripped leading/trailing whitespaces from column names for accurate referencing.
 - Handled missing values by removing null rows.
 - Converted the 'Date' column from a string to a datetime object to facilitate time-series plotting.
- **Exploratory Data Analysis (EDA):**
 - Calculated mean unemployment rates for "Pre-Covid" (before April 2020) and "Post-Covid" periods.
 - Grouped data by 'Region' and 'Area' to identify geographical and sectoral disparities.
- **Visualization:**
 - **Line Plots:** To track the unemployment trend over time.
 - **Bar Charts:** To compare the average unemployment rates across different Indian states.

5. Key Findings & Insights

A. The Pandemic Spike

The analysis confirms a sharp increase in joblessness following the pandemic's arrival:

- **Pre-Covid Average:** ~9.61%.
- **Post-Covid Average:** ~20.19%. The unemployment rate more than **doubled** during the lockdown phase.

B. Sectoral Comparison (Urban vs. Rural)

- Urban areas consistently reported higher unemployment rates compared to Rural areas throughout the period.
- Both sectors saw a simultaneous spike in April 2020, but the recovery (downward trend) began slightly earlier in Rural regions.

C. Regional Impact

The economic shock was not uniform across India:

- **Highest Unemployment:** States like **Tripura, Haryana, and Jharkhand** recorded the highest average rates.
- **Lowest Unemployment:** States like **Meghalaya, Odisha, and Assam** remained relatively more stable.
- **Extreme Values:** A peak unemployment rate of **76.74%** was recorded in certain urban pockets during the height of the lockdown.

7. Conclusion

The analysis successfully highlights the vulnerability of the labor market to large-scale external shocks. The "Unemployment in India" project demonstrates that while the pandemic was a national crisis, its impact was localized and sector-specific, with urban centers bearing the brunt of the economic shutdown. This project serves as a foundational step for further predictive modeling or policy-impact studies.